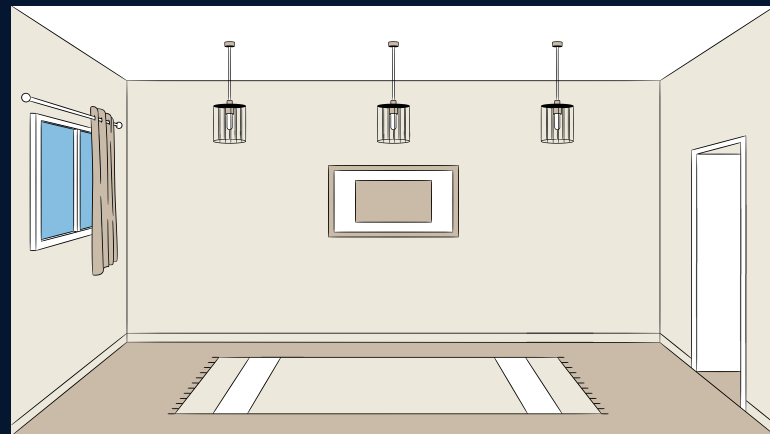


# HTML

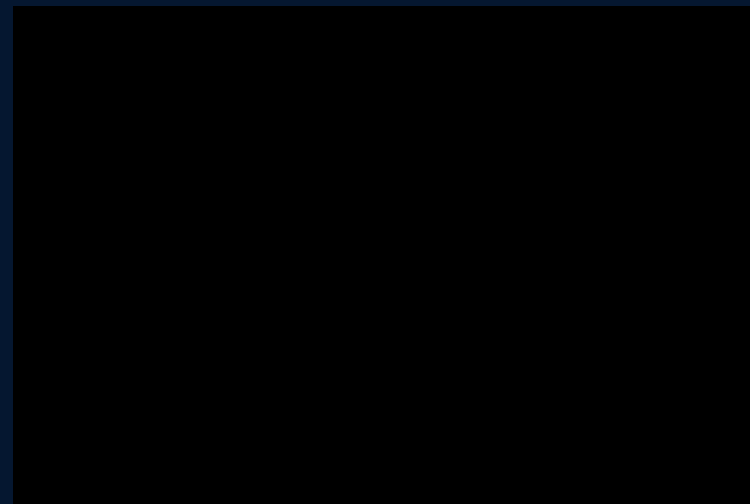
# CSS

# JS

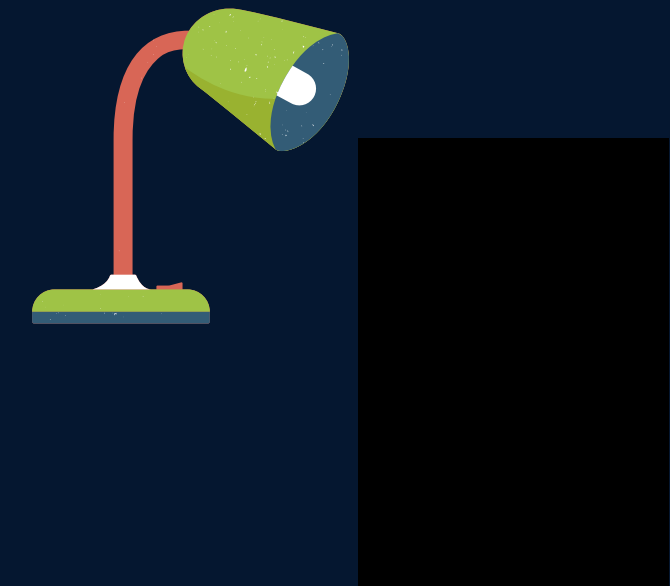
## Structure/layout



## Style



## Logic



# Level 1



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# CSS

## Cascading Style Sheet

*It is a language that is used to describe the **style** of a document.*

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# Basic Syntax

Selector

**h1 {**

**color: red;**

**}**

Property

Value

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# Including Style

- Inline

```
<h1 style="color: red"> Apna College </h1>
```

- <style> tag

```
<style>  
  h1 {  
    color : red;  
  }  
</style>
```



# Including Style

- External Stylesheet

Writing CSS in a separate document & linking it with HTML file

# Color Property

*Used to set the color of foreground*

color: red;

color: pink;

color: blue;

color: green;

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# Background Color Property

*Used to set the color of **background***

background-color: **red**;

background-color: **pink**;

background-color: **blue**;

background-color: **green**;



# Color Systems

- RGB

color: `rgb(255, 0, 0);`

color: `rgb(0, 255, 0);`



# Color Systems



- Hex (Hexadecimal)

color: `#ff0000`;

color: `#00ff00`;

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# Selectors

- Universal Selector

```
* { }
```

- Class Selector

```
.myClass { }
```

- Element Selector

```
h1 { }
```

- Id Selector

```
#myId { }
```

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# Practice Set 1

Q1: Create a simple div with an id "box".

Add some text content inside the div.

Set its background color to blue.

Q2: Create 3 headings with h1, h2 & h3.

Give them all a class "heading" & set color of "heading" to red.

Q3: Create a button & set its background color to :

- green using css stylesheet
- blue using <style> tag
- pink using inline style

# Text Properties

text-align

text-align : left / right / center

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# Text Properties

text-decoration

text-decoration : underline / overline / line-through

# Text Properties

font-weight

font-weight : normal / bold / bolder / lighter

font-weight : 100-900

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# Text Properties

font-family

font-family : arial

font-family : arial, roboto

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# Units in CSS

## *Absolute*

pixels (px)

96px = 1 inch

font-size: 2px;

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# Text Properties

line-height

line-height : 2px

line-height : 3

line-height : normal

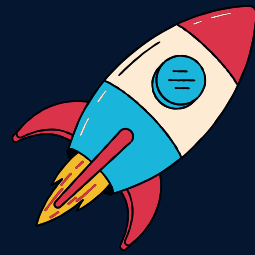
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# Text Properties

text-transform

text-transform : uppercase / lowercase / capitalize / none

## Practice Set 2



Q1: Create a heading centred on the page with all of its text capitalized by default.

Q2: Set the font family of all the content in the document to "Times New Roman".

Q3: Create one div inside another div.

Set id & text "outer" for the first one & "inner" for the second one.

Set the outer div text size to 25px & inner div text size to 10px.



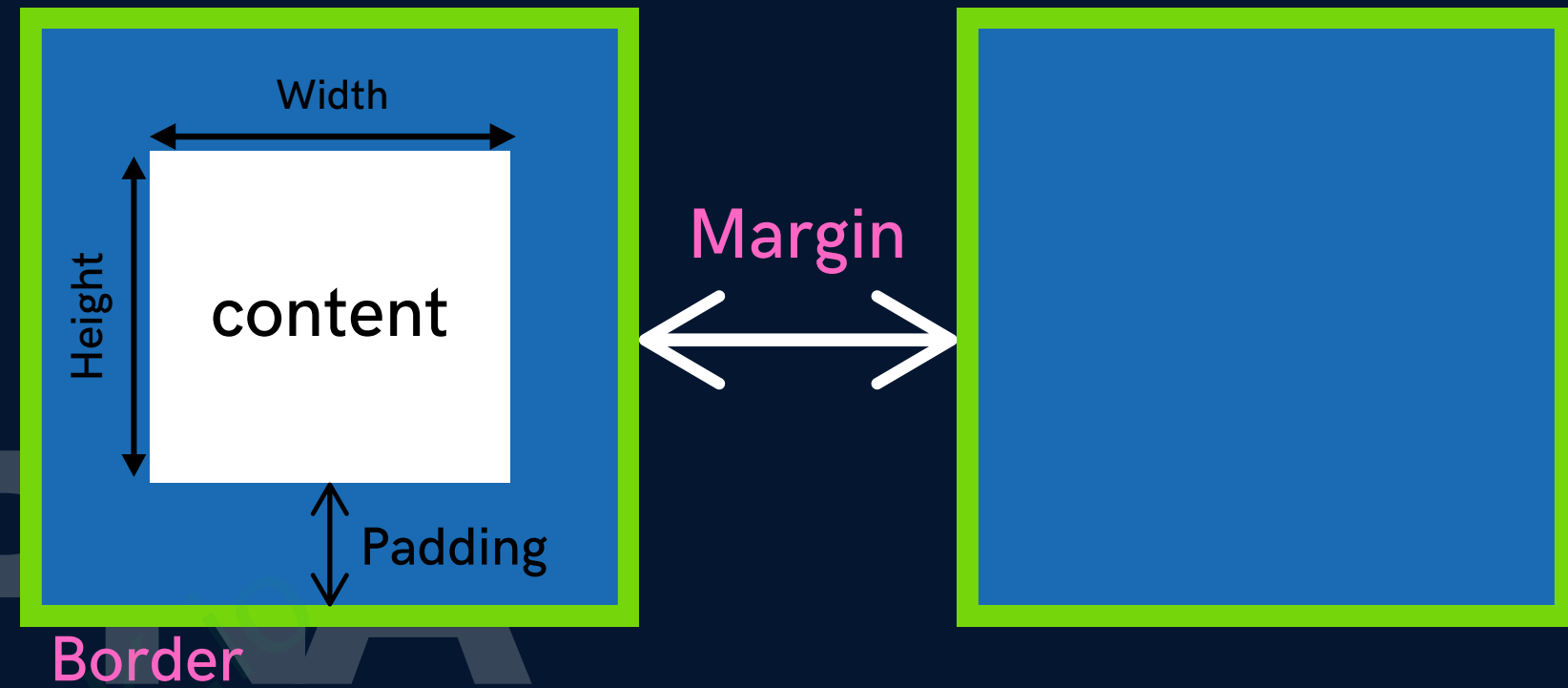
# Level 2



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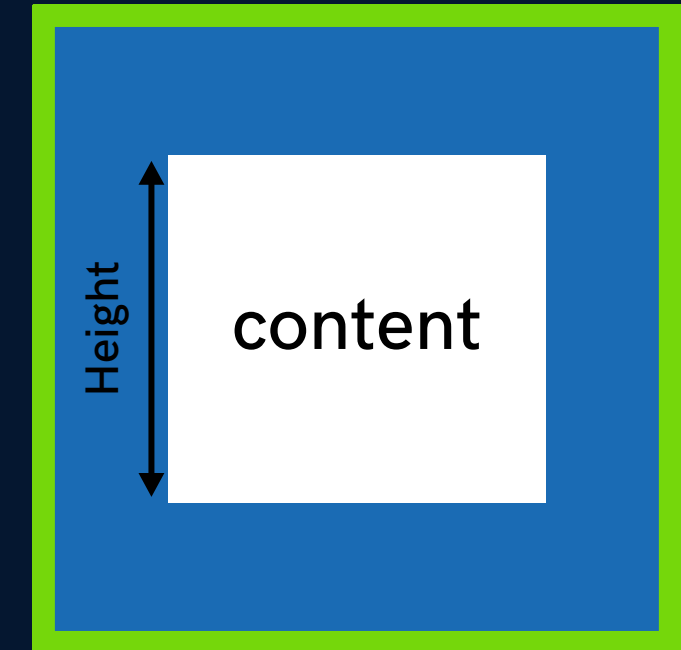
# Box Model in CSS

- Height
- Width
- Border
- Padding
- Margin



# Height

By default, it sets the content area **height** of the element



```
div {
```

```
    height: 50px;
```

```
}
```

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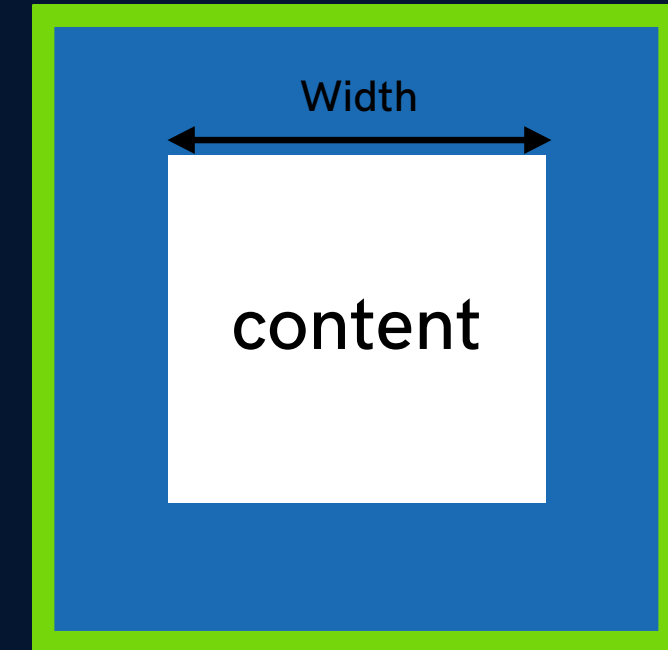
# Width

By default, it sets the content area **width** of the element

```
div {
```

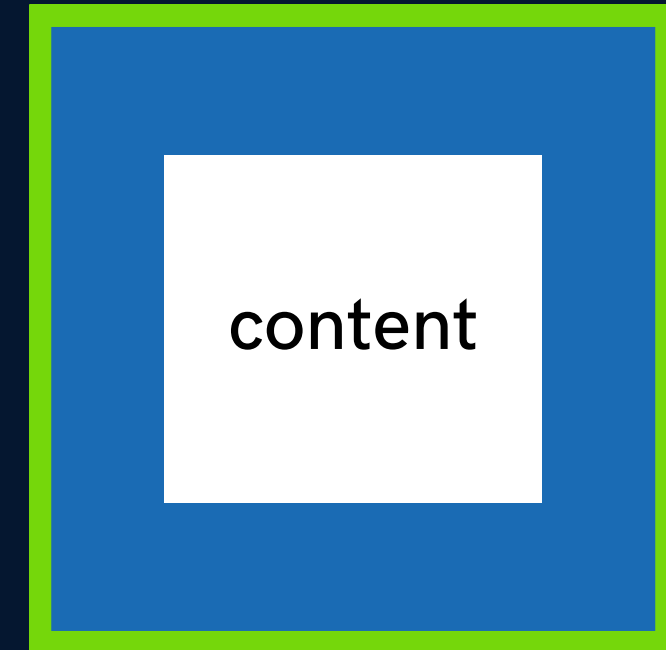
```
width: 50px;
```

```
}
```



# Border

Used to set an element's **border**



Border

*border-width : 2px;*

*border-style : solid / dotted / dashed*

*border-color : black;*

# Border

## Shorthand

*border : 2px solid black;*

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# Border

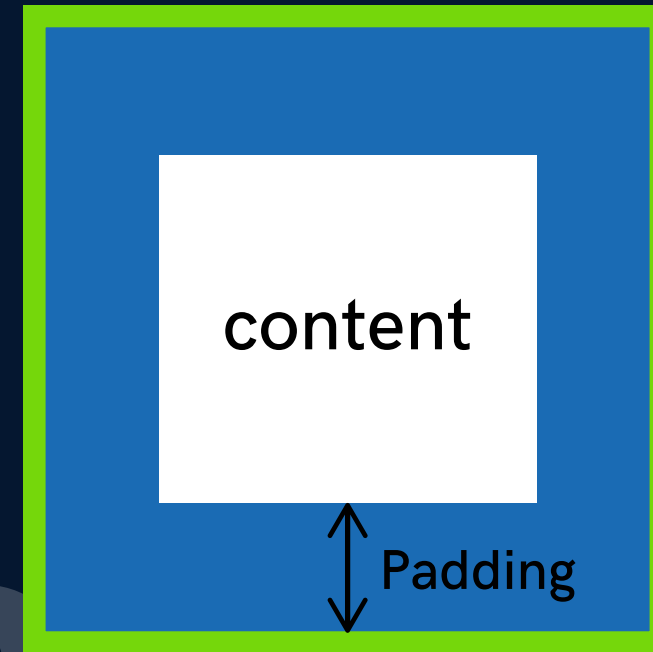
Used to **round the corners** of an element's outer border edge

*border-radius : 10px;*

*border-radius : 50%;*

# Padding

- padding-left
- padding-right
- padding-top
- padding-bottom



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# Padding

## Shorthand

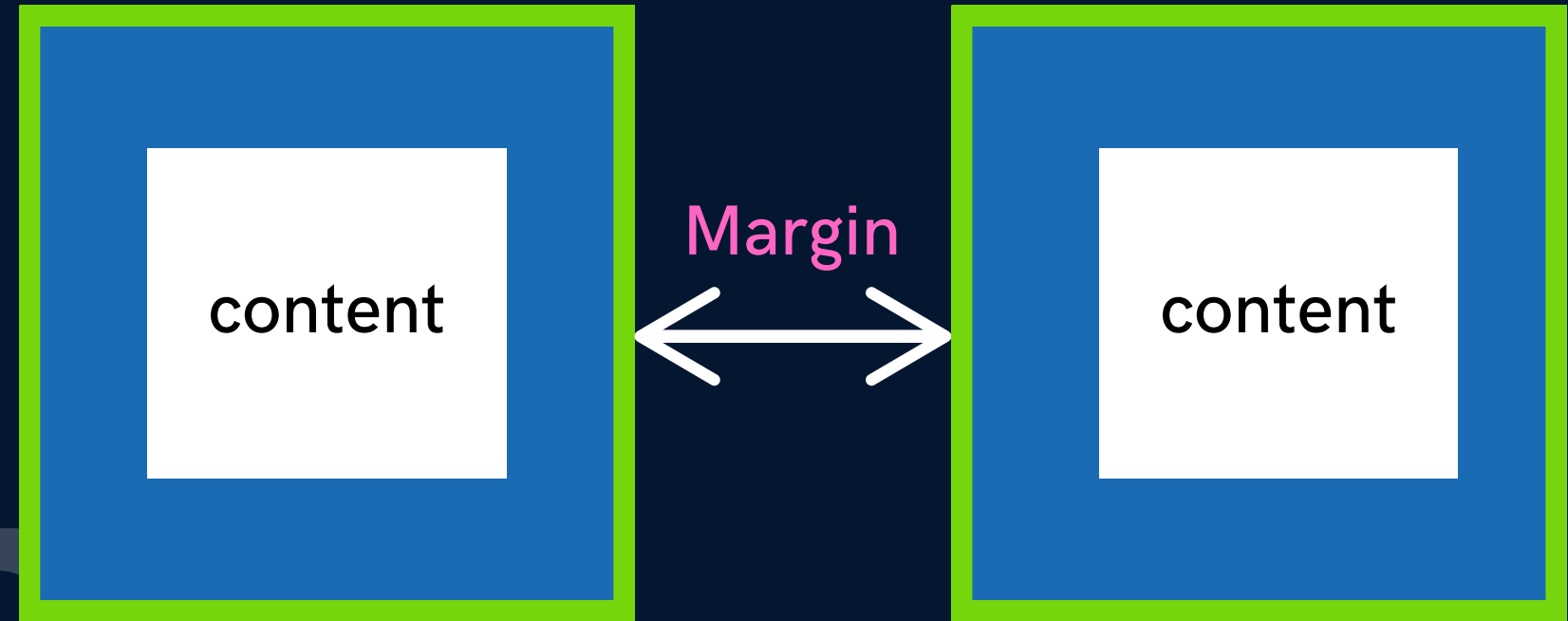
*padding: 50px;*

*padding: 1px 2px 3px 4px;*

top | right | bottom | left -> clockwise

# Margin

- margin-right
- margin-left
- margin-top
- margin-bottom



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# Margin

## Shorthand

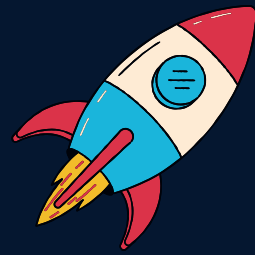
*margin: 50px;*

*margin: 1px 2px 3px 4px;*

top | right | bottom | left -> clockwise



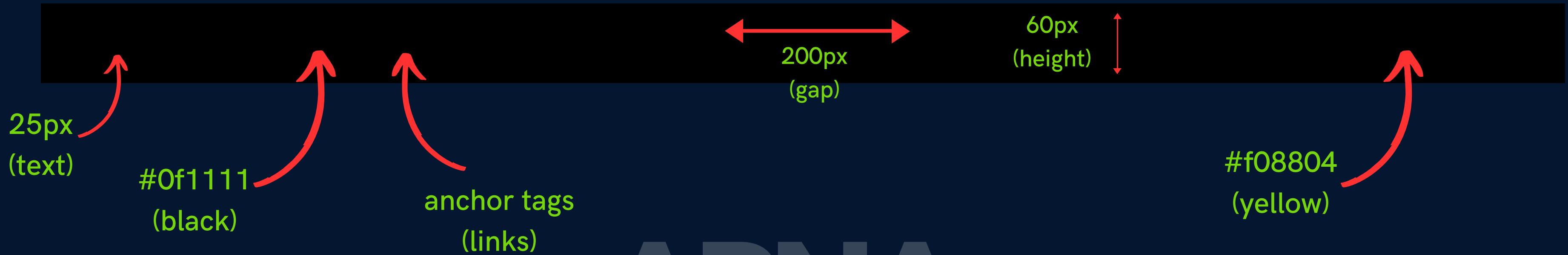
## Practice Set 3



Q1: Create a div with height & width of 100px.  
Set its background color to green & the border radius to 50%.

Q2: Create the following navbar.

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# Display Property

*display: inline / block / inline-block / none*

- **inline** - Takes only the space required by the element. (no margin/ padding)
- **block** - Takes full space available in width.
- **inline-block** - Similar to inline but we can set margin & padding.
- **none** - To remove element from document flow.

# Visibility

*visibility: hidden;*

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**Note** : When visibility is set to none, space for the element is reserved.

But for display set to none, no space is reserved or blocked for the element.

# Alpha Channel

opacity (0 to 1)

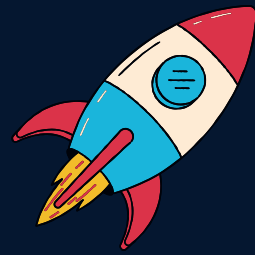
- RGBA

color: `rgba(255, 0, 0, 0.5);`

color: `rgba(255, 0, 0, 1);`

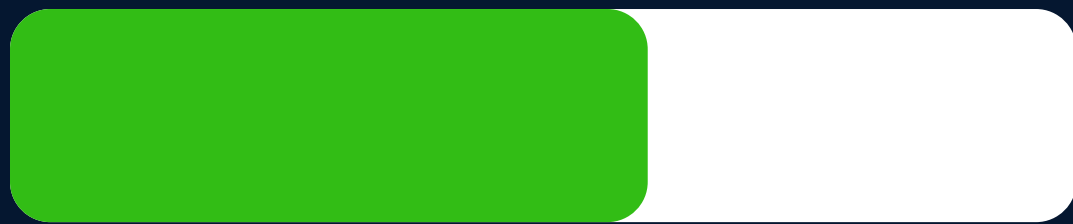


## Practice Set 4



- Q1: Create a webpage layout with a header, a footer & a content area containing 3 divs.  
Set the height & width of divs to 100px.  
(add the previous navbar in the header)
- Q2: Add borders to all the divs.
- Q3: Add a different background color to each div with an opacity of 0.5
- Q4: Give the content area an appropriate height.

# Level 3



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# Units in CSS

## *Relative*

%

em

rem

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# Percentage (%)

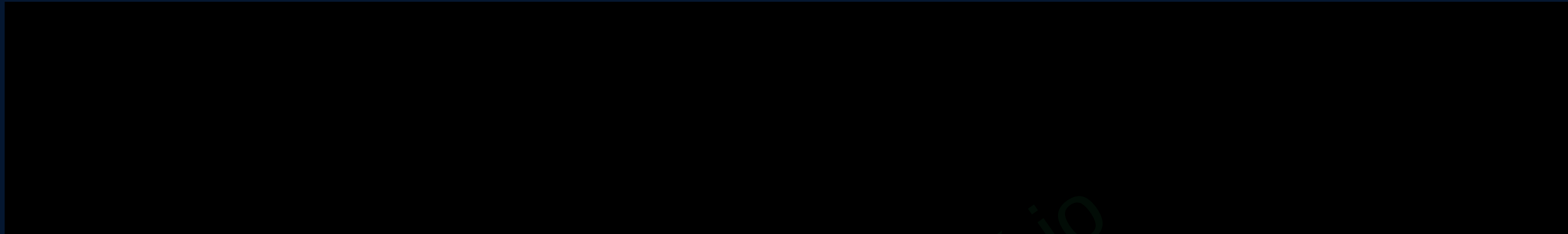
It is often used to define a size as relative to an element's parent object.

*width : 33% ;*

*margin-left : 50% ;*

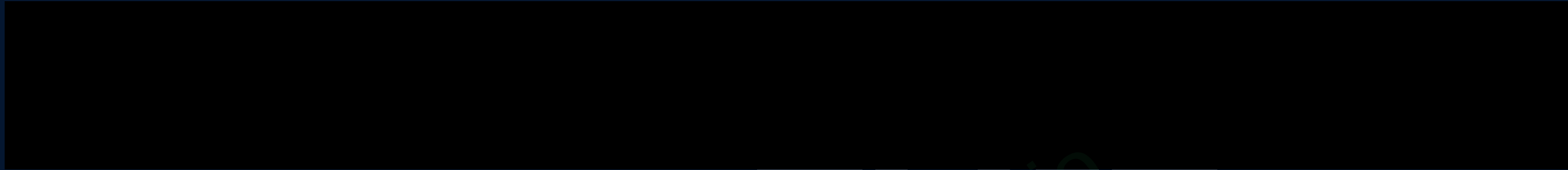
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Em



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# Rem (Root Em)



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# Others

vh: relative to 1% viewport height

vw : relative to 1% viewport width

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# Position

The position CSS property sets **how an element is positioned** in a document.

*position : static / relative / absolute / fixed*

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# Position

- **static** - default position (The top, right, bottom, left, and z-index properties have no effect)
- **relative** - element is relative to itself. (The top, right, bottom, left, and z-index will work)
- **absolute** - positioned relative to its closest positioned ancestor. (removed from the flow)
- **fixed** - positioned relative to browser. (removed from flow)
- **sticky** - positioned based on user's scroll position

# z-index

It decides the **stack level** of elements

Overlapping elements with a larger z-index cover those with a smaller one.

*z-index : auto (0)*

*z-index : 1 / 2 / ...*

*z-index : -1 / -2 / ...*

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# Background Image

Used to set an image as background

```
background-image : url("image.jpeg");
```

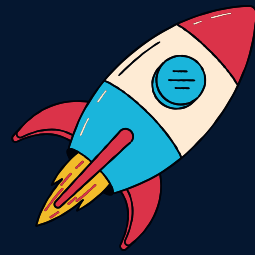


# Background Size

*background-size : cover / contain / auto*

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# Practice Set 5



Qs: Create the following layout using the given html.

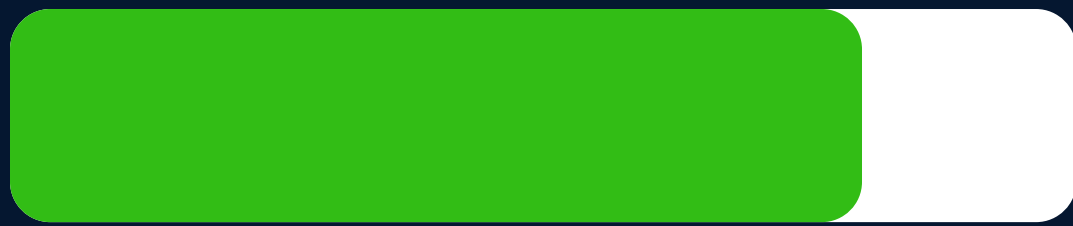
- Give the div a height, width & some background image.
- Use the appropriate position property for the div element to place it at the right end of the page. (The div should not move even on scroll)
- Use z-index to place the div on top of page.

```
<p> lorem*5 </p>
```

```
<div> Love Nature </div>
```

```
<p> lorem*5 </p>
```

# Level 4



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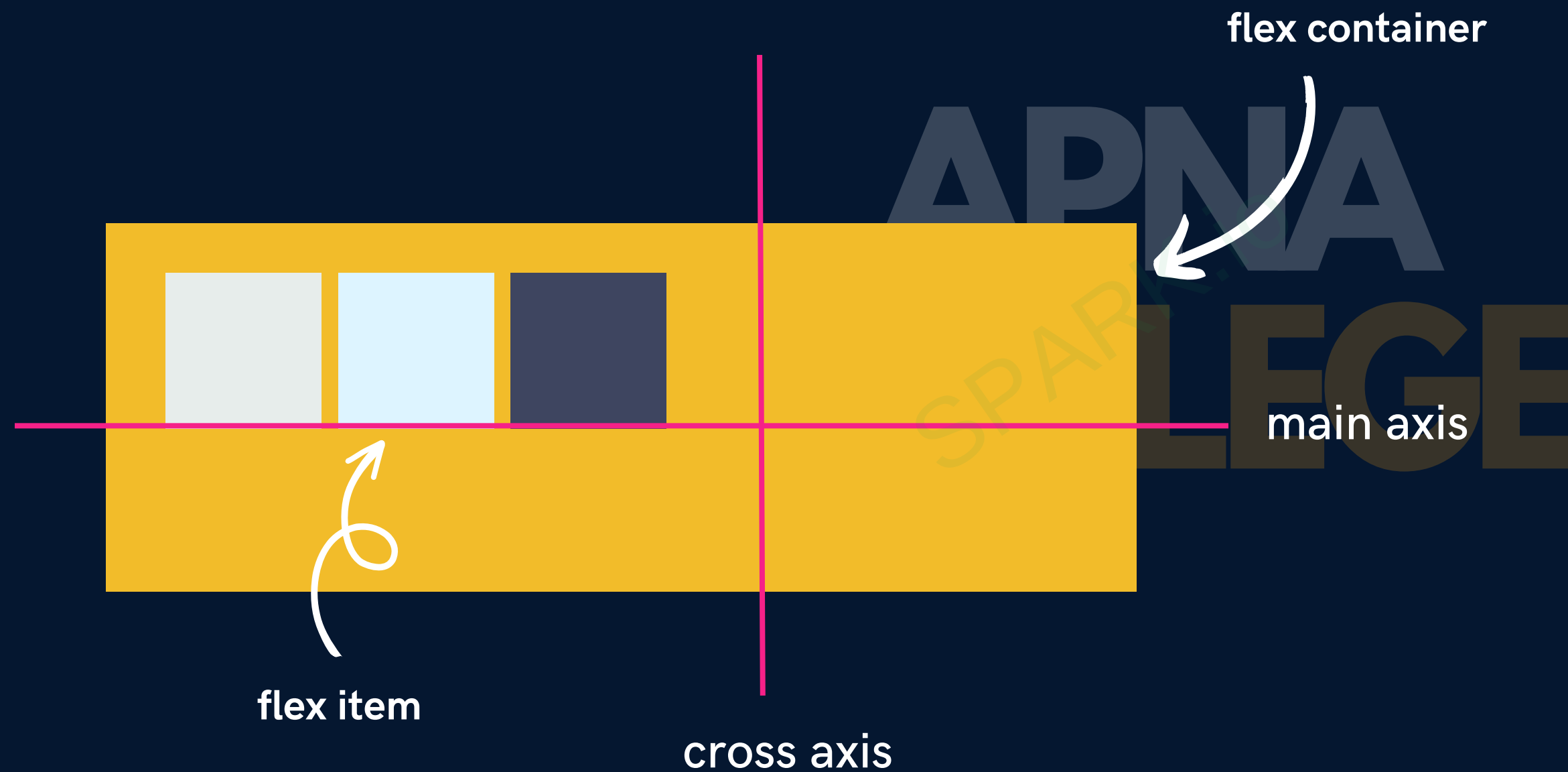
# Flexbox

## Flexible Box Layout

It is a one-dimensional layout method for arranging items in rows or columns.

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# The Flex Model





# Flexbox Direction

It sets how flex items are placed in the flex container, along which axis and direction.

- `flex-direction : row;` (default)
- `flex-direction : row-reverse;`
- `flex-direction : column;`
- `flex-direction : column-reverse;`

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# Flex Properties

for Flex Container

- **justify-content** : alignment along the main axis.  
flex-start / flex-end / centre / space-evenly /
- **flex-wrap** : nowrap / wrap / wrap-reverse
- **align-items** : alignment along the cross axis.
- **align-content** : alignment of space between & around the content along cross-axis

# Flex Properties

for Flex Item

- **align-self** : alignment of individual along the cross axis.
- **flex-grow** : how much a flex item will grow relative to the rest of the flex items if space is available
- **flex-shrink** : how much a flex item will shrink relative to the rest of the flex items if space is available



## Practice Set 6

Qs: Create a navbar with 4 options in the form of anchor tags inside list items.  
Now, use flexbox to place them all spaced equally in a single line.

Qs: Use flexbox to center one div inside another div.

Qs: Which has higher priority - align-items or align-self?

# Media Queries

Help create a responsive website

```
@media (width : 600px) {  
  div {  
    background-color : red;  
  }  
}
```

```
@media (min-width : 600px) {  
  div {  
    background-color : red;  
  }  
}
```

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# Media Queries

```
@media (min-width : 200px) and (min-width : 300px) {  
  div {  
    background-color : red;  
  }  
}
```

# Practice Set 7

Qs: Add a media query to implement the following:

- the color of a div changes to green for viewport width less than 300px
- the color of a div changes to pink for width between 300px & 400px
- the color of a div changes to red for width between 400px & 600px
- the color of a div changes to blue for width above 600px

# Level 5



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# Transitions

Transitions enable you to define the transition between two states of an element.

- `transition-property` : property you want to transition (font-size, width etc.)
- `transition-duration` : 2s / 4ms ..
- `transition-timing-function` : ease-in / ease-out / linear / steps ..
- `transition-delay` : 2s / 4ms ..

# Transition Shorthand

property name | duration | timing-function | delay

*transition: font-size 2s ease-in-out 0.2s;*

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# CSS Transform

Used to apply 2D & 3D transformations to an element

- rotate

*transform: rotate(45deg);*

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# CSS Transform

- scale

*transform: scale(2);*

*transform: scale(0.5);*

*transform: scale(1, 2);*

*transform: scaleX(0.5);*

*transform: scaleY(0.5);*

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# CSS Transform

- translate

*transform: translate(20px);*

*transform: translate(20px, 50px);*

*transform: translateX(20px);*

*transform: translateY(20px);*

# CSS Transform

- skew

*transform: skew (30deg);*

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# Animation

To animate CSS elements

```
@keyframe myName {  
  from { font-size : 20px; }  
  to { font-size : 40px; }  
}
```

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# Animation Properties

- animation-name
- animation-duration
- animation-timing-function
- animation-delay
- animation-iteration-count
- animation-direction

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# Animation Shorthand

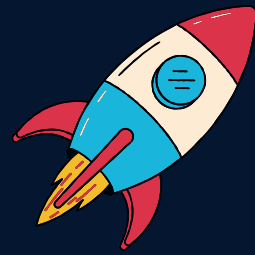
*animation : myName 2s linear 3s infinite normal*

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# % in Animation

```
@keyframe myName {  
  0% { font-size : 20px; }  
  50% { font-size : 30px; }  
  100% { font-size : 40px; }  
}
```

# Practice Set 8



Qs: Create a simple loader using CSS

**Step1** : create a div with circular shape & a thick border from one end (top/bottom/left/right)

**Step2** : To make it spin create an animation which transforms it from 0deg to 360deg

**Step3** : Add the animation property to the loader with infinite duration